

GCP regression

Stage 2: Sparse/GCP anchoring:

Possibility A: If giving the model just 3 to 5 known height points dramatically fixes the whole image, it means DAv2 is actually doing a great job at figuring out the shape of the city. It just needs a "local ruler" to tell it how big things are in that specific frame.

Possibility B: If giving it 3 to 5 known points still results in massive errors, it means DAv2 is actively hallucinating or distorting the actual shape of the buildings. A local ruler won't fix a warped map. If this happens, it proves you absolutely must use a heavier AI model (like RDAH-Net) to fix the structural errors.

For DFC2019, sparse GCP coordinates/heights for these tiles are not available.

Presenting "synthetic sparse metric anchors or simulated sparse ground-control measurements derived from reference AGL."

In scripts, created **evaluate_sparse_anchors.py**: This sparse anchor evaluator picks a tiny number of pixels from the test tile, pretending their height as the only metric measurements available, fit a local DAv2→height mapping from them, then evaluate the mapping on the rest of the tile. We'll test 2, 3, 5, 10, 20 anchors. Anchor selection criteria: spatially separated and vertically diverse. Evaluation criteria: The anchor pixels themselves are excluded from the evaluation.

Results:

data/dfc2019/experiments/sparse_anchor/

- ├ sparse_anchor_metrics.csv
- ├ sparse_anchor_summary.csv
- └ sparse_anchor_city_summary.csv

Inference: Our global isotonic baseline was:

Metric	Value
MAE	3.691 m
RMSE	5.313 m
Pearson	0.556
Spearman	0.446

Metric	Value
MAE	4.012 m
RMSE	5.197 m
Pearson	0.492
Spearman	0.412

Even 20 simulated anchors only gives:

MAE 4.012 m

RMSE 5.197 m

Pearson 0.492

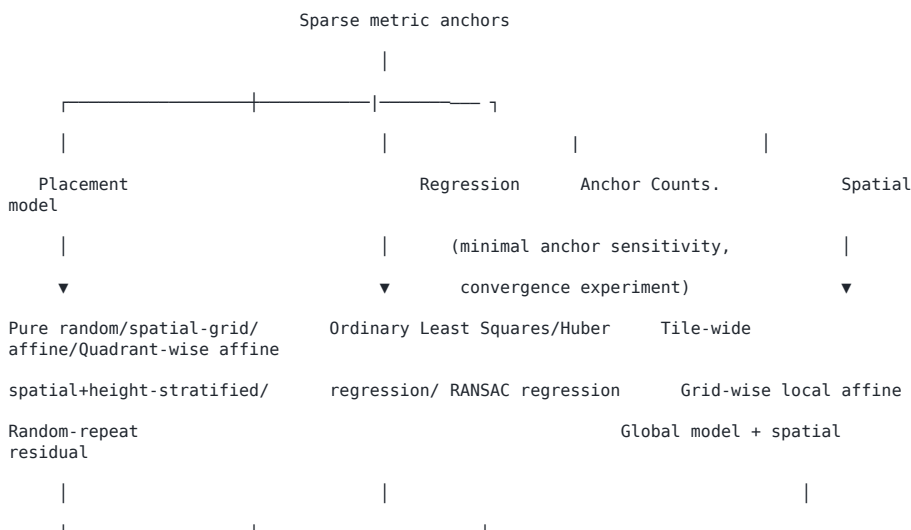
Spearman 0.412

So this particular sparse-anchor implementation does not beat the global isotonic baseline.

Hypothetical Reasoning:

We are fitting: $AGL = a \times DAv2 + b$ using anchors from the same test tile, and then evaluating the rest of that tile. That's okay for a simulated local-calibration experiment. But real GCP-based reconstruction would normally involve: carefully surveyed metric points, spatial distribution chosen for geometric coverage, robust regression, outlier handling, possibly piecewise/spatial correction. Our tiny affine fit is only a very simple baseline. We decided to revisit the sparse-anchor experiment.

Turning our experiment into a four-stage study:



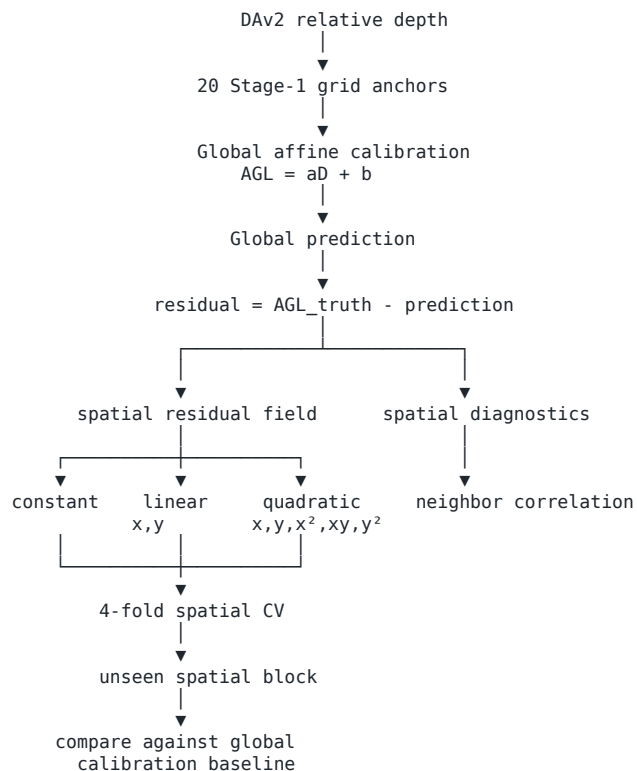
Quadrant-wise	3.356 m	5.515 m	0.459	0.389
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Introducing local independent spatial models made things worse. That is a very useful negative result.

Stage 4 —

Test whether the remaining residual error is spatially structured without splitting the calibration data into independent local regressions.

We would compare: A. Global tile-wide affine against B. Global affine + LOW-COMPLEXITY spatial residual, with the residual model deliberately constrained.



A: Global tile wide affine
In scripts, we created
evaluate_sparse_anchor_spatial_residual.py

Result:

data/dfc2019/experiments/sparse_anchor_spatial_residual/

```

├── config.json
├── preflight.csv
├── results.csv
├── summary.csv
├── tile_summary.csv
├── paired_deltas.csv
├── spatial_diagnostics.csv
├── oracle_in_sample.csv
├── REPORT.txt
├── diagnostics/
├── JAX_004_006.npz
├── JAX_004_014.npz
├── ...

```

	Result
constant	✗
global affine	baseline
linear x,y	✓ modest improvement
quadratic x,y	✗ overfits / generalizes worse

The error has spatial structure, but it's not well described by a low-order polynomial.

The current Stage 4 is explicitly an offline oracle experiment: the full AGL truth is used to train the residual model. The report itself correctly says that a positive result demonstrates residual learnability, not deployment feasibility.

Stage 4B — regularized smooth residual field

20 grid anchors

↓

global OLS calibration

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$\text{residual} = \text{truth} - \text{global prediction}$

↓

regularized smooth spatial model

↓

4-fold spatial blocked CV

↓

compare against global baseline

In scripts, created **evaluate_sparse_anchor_smooth_residual.py**

Find results in:

DepthWizard2/data/dfc2019/experiments/sparse_anchor_smooth_residual

The paired analysis is brutal:

- rbf_medium improves RMSE on 0/50 tiles
- rbf_strong improves RMSE on 0/50
- rbf_weak improves RMSE on 0/50
- Even MAE improves on only 2/50, 1/50, and 1/50 tiles respectively.

Stage 4A found:

the residual is highly spatially correlated and a simple linear spatial trend gives a modest improvement.

Stage 4B found:

trying to model that residual with a flexible smooth interpolator generalizes horribly.

The residual has spatial structure, but it is not behaving like a smoothly interpolatable scalar surface that can be reconstructed from coordinate position alone.

New baseline: Global 20-anchor grid calibration